Analysis of Olist Shoppers in SP with Spectral Co-Clustering

1 Overview

This section builds upon the findings from the Olist descriptive analytics case study, with a summary available here. In this example, we illustrate a second common theme in the application of graph-based analysis in machine learning: modeling data as a bipartite graph to uncover meaningful patterns and relationships.[1]. The bipartite graph model finds extensive application, for a comprehensive review, see [3]. This example tells you how shoppers used Olist in Sau Paulo in the year 2017. In particular, it brings out the patterns in shopping activity over the weeks of the year in 2017.

2 Data Description

The dataset originates from Kaggle and contains transaction records from a Brazil-based online retailer. A review of the links in the previous section would show that the year 2017 offers the most complete and representative sample of shopping activity, with São Paulo accounting for nearly 40% of all transactions. Analyzing this major market is therefore crucial. This analysis looks at shopping activity in Sau Paulo during 2017 notebook. In this study, we extend the analysis using a co-clustering approach, which simultaneously groups both weekly purchase activity and products purchased. This method reveals affinities between specific products and particular periods (weeks) of the year.

3 Methodology

1. Begin with the prepared dataset from the descriptive analytics exercise: freq_prod_weekly_sale_SP_2017.parquet.
2. Model the data as a bipartite graph, where one vertex set represents the weeks of the year and the other represents product inventory. An edge between a week and a product indicates that the product was sold during that week (i.e., a nonzero entry in the data matrix).
3. Prepare the data for the spectral co-clustering algorithm described in [2], including computation of the normalized Laplacian.

[1] See the documents under the sba loans descriptive analysis, for an example of using homogeneous graphs in machine learning and analytics applications.Homogeneous graphs are the first common theme in application of graph in machine learning and analytics

4. Apply co-clustering, which simultaneously groups both rows (weeks) and columns (products) to uncover subsets that exhibit similar patterns. This approach provides a more granular view than traditional clustering, revealing associations between specific products and time periods. The methodology is similar to the spectral clustering approach used in the risky borrowers analysis, with adaptations for bipartite graphs as detailed in Section 4.

5. Determine the optimal number of clusters using spectral gap analysis, following the procedure from the sba risky borrowers example.

6. Interpret and profile the resulting co-clusters to extract actionable insights.

4 Spectral Co-Clustering

This section outlines the conceptual framework behind spectral co-clustering, following the methodology of [2]. Unlike traditional clustering, which groups either rows (examples) or columns (features) independently, co-clustering simultaneously identifies subsets of both rows and columns that exhibit similar patterns. This approach is particularly effective for high-dimensional datasets, such as gene expression data, text corpora, or medical imaging, where meaningful associations often exist between specific groups of features and data points.

The core idea is the duality between clustering rows and columns: clusters of examples often correspond to clusters of features, and vice versa. In bipartite data, this relationship is captured by a rectangular data matrix, rather than the square similarity matrices used in standard spectral clustering. The algorithm proceeds by constructing a normalized Laplacian from the bipartite graph, then analyzing its eigenvalues to determine the number of clusters via the spectral gap.

The embedding for clustering is derived from the left and right eigenvectors of the Laplacian, representing the two vertex sets (weeks and products). Following [2], the number of embedding dimensions l is chosen as $l = \lceil \log k \rceil$, where k is the number of clusters identified. K-Means is then applied to these embeddings to obtain the final co-clusters.

For a comprehensive review of co-clustering methods, see [1]. Implementation details are available in this notebook.

5 Results

To set up the discussion of the results, let's start with what we know from the results of conventional clustering on this data, available here. The neighborhood graph was set up with a cosine similarity measure. A threshold of 0.4 was used to compute the neighborhood graph. Nodes with cosine similarity greater than this threshold are connected. By inspecting the distance matrix visually (using a heatmap), four clusters were found. Spectral clustering of this cosine similarity graph produced the clustering shown in. figure 5. We see one big cluster and three smaller clusters corresponding to what we think are either sales promotion or holiday season based activity.

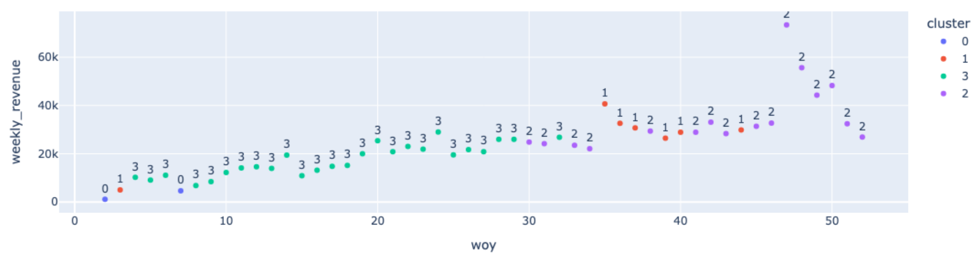

Figure 1: Spectral clustering with cosine similarity on SP 2017 weekly shopping data. The figure shows the clusters identified among weeks based on shopping patterns, using a cosine similarity threshold of 0.4.

In the conventional clustering approach, the number of clusters was determined by visually inspecting the heatmap, rather than using the spectral gap method. As will be shown, both approaches yield similar results for this dataset. However, a key limitation of traditional clustering is that it does not directly reveal associations between products and clusterssuch insights require additional post-hoc analysis. In contrast, co-clustering simultaneously identifies groups of weeks and products, providing these associations as part of the clustering output.

To set up the co-clustering algorithm, we compute the *normalized laplacian* matrix of the shopping activity for 2017. We run *SVD* on it and plot the (ordered) eigenvalues. This is shown in figure 2

For this data, the optimal number of clusters (k), is 16. We provide this as an input to the co-clustering algorithm and the algorithm reports the row (week of the year) and column (product) clusters. These are shown in figure 3 and figure 4.

A review of the *week* component in figure 3 shows a large cluster of 46 weeks and some small clusters. This is consistent with what is observed in the results of similarity clustering, figure 5. With co-clustering, we can pick the *column* components of a cluster directly. For example, the product-id's in *cluster-8* of the product clusters is shown in figure 5. This additional resolution is a feature of co-clustering.

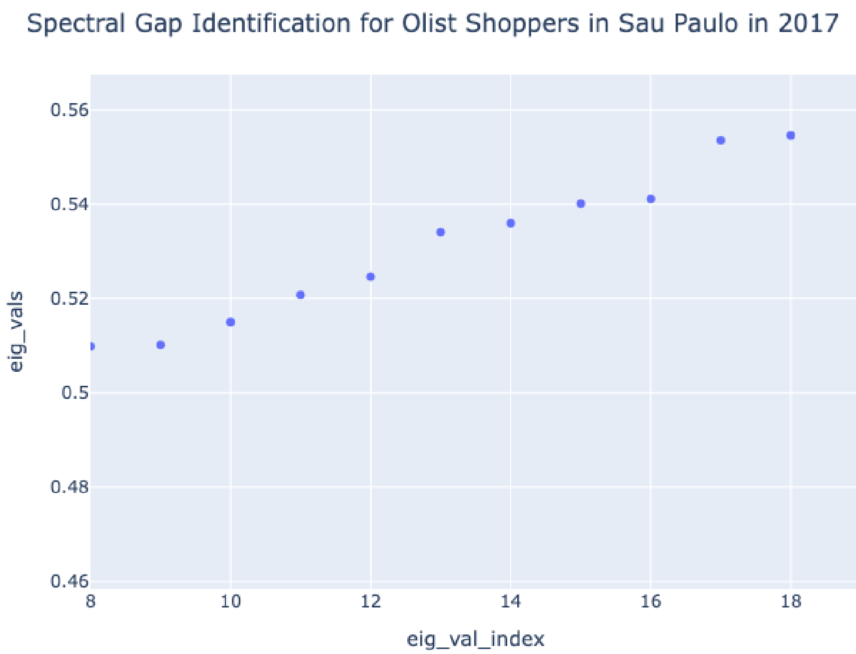

Figure 2: Spectral gap identification for co-clustering. The plot shows the ordered eigenvalues of the normalized Laplacian, used to determine the optimal number of clusters.

cluster	size
0	46
3	2
12	2
6	1
8	1

Figure 3: Weekly activity clusters

	cluster	size
0	0	2794
1	8	40
2	12	22
3	5	19
4	7	17
5	2	16
6	3	13
7	1	12
8	14	5
9	10	5
10	13	5
11	9	5
12	15	4
13	11	4
14	4	4

Figure 4: Product purchase clusters

6 Discussion and Insights

Spectral co-clustering provides actionable insights into complex, high-dimensional datasets by simultaneously grouping rows and columnshere, weeks and products. This dual clustering approach enables practitioners to not only identify temporal patterns in shopping activity, but also to directly associate

cluster products
0705b33fb285827a578fce4899f1b921
0f0c4960b05d6145bbb4b348423503ef
222efa72a47277d611b8b38d71149afd
2635c3e7db0ac6cb3e733cac61ce0ba5
2d2600c283c0c1af0528fd84c14a2c28
2eb9b2ef7c1da3c7b99702452ea4729f
336cfb12a5468070795ae4c964f4ce47
419538c5c0c21d892928586bee1ea16a
45a15b38cc3c0514717a1de673c6193c
4bb4fb9c85785b75ab4f6559900c7ca1

Figure 5: Product cluster 8 partial composition (product ID's)

specific products with distinct periods of heightened or specialized demand. For example, business analysts can pinpoint which products are characteristic of holiday or promotional clusters, while researchers in other domainssuch as healthcare or neurosciencecan use co-clustering to reveal associations between groups of features (gene expression or EEG signal) and data points (patients or subjects), leading to more interpretable and targeted findings. In this example, the method uncovers both the overall temporal aggregation of shopping activity in São Paulo during 2017 and the product categories that define each cluster, offering a richer understanding than conventional clustering approaches.

References

[1] Elena Battaglia, Federico Peiretti, and Ruggero Gaetano Pensa. "Co-clustering: A Survey of the Main Methods, Recent Trends, and Open Problems". In: *ACM Comput. Surv.* 57.2 (Nov. 2024). ISSN: 0360-0300. DOI: 10.1145/3698875. URL: https://doi.org/10.1145/3698875.

[2] Inderjit S Dhillon. "Co-clustering documents and words using bipartite spectral graph partitioning". In: *Proceedings of the seventh ACM SIGKDD international conference on Knowledge discovery and data mining*. 2001, pp. 269–274.

[3] Jérôme Kunegis. "Exploiting the structure of bipartite graphs for algebraic and spectral graph theory applications". In: *Internet Mathematics* 11.3 (2015), pp. 201–321.